CLCSA16 DWDM

CLASSROOM ACTIVITY

SHORT ANSWER TEST

TOPIC: Classification of Data Mining Systems **Time:30 mins**

S. No.	Question	BTL Term	BTL Level
1	Define a Data Mining System.	Remember	BTL1
2	List the different classifications of Data Mining Systems.	Remember	BTL1
3	Explain the classification of Data Mining Systems based on the type of data mined.	Understand	BTL2
4	Describe the classification of Data Mining Systems based on the type of knowledge mined.	Understand	BTL2
5	Identify the techniques used in Data Mining Systems.	Remember	BTL1
6	Explain the classification of Data Mining Systems based on applications.	Understand	BTL2
7	Describe the classification of Data Mining Systems based on user interaction.	Understand	BTL2
8	Illustrate a real-world application of a Data Mining System in healthcare.	Apply	BTL3
9	Define classification in data mining.	Remember	BTL1
10	Differentiate between classification and clustering in data mining.	Understand	BTL2

BTL = Bloom's Taxonomy Level

Answers

1. Define a Data Mining System.

A Data Mining System is a computer-based system used to extract useful information, patterns, and knowledge from large amounts of data. It helps users discover hidden relationships and make better decisions.

2. List the different classifications of Data Mining Systems.

Data Mining Systems are classified based on:

1. Type of data mined.
2. Type of knowledge mined.
3. Techniques used.
4. Applications.
5. User interaction.

3. Explain the classification of Data Mining Systems based on the type of data mined.

Based on the type of data mined, Data Mining Systems are classified as follows:

1. Relational databases: Mine data stored in tables containing rows and columns.
2. Transactional databases: Analyze customer transactions and purchase records.
3. Data warehouses: Extract useful patterns from integrated historical data.
4. Spatial databases: Analyze geographical and location-based data.
5. Time-series databases: Identify trends and patterns over time.
6. Web data: Extract information from websites, web pages, and online activities.

4. Describe the classification of Data Mining Systems based on the type of knowledge mined.

Based on the type of knowledge mined, Data Mining Systems include:

1. Association: Finds relationships between items, such as products purchased together.

2. Classification: Assigns data to predefined categories.
3. Clustering: Groups similar data objects without predefined categories.
4. Prediction: Predicts future values or outcomes using existing data.
5. Outlier analysis: Identifies unusual data that differs from normal patterns.
6. Trend analysis: Identifies changes and patterns over time.

5. Identify the techniques used in Data Mining Systems.

The major techniques used in Data Mining Systems are:

1. Classification
2. Clustering
3. Association rule mining
4. Regression
5. Prediction
6. Outlier detection
7. Sequential pattern mining

6. Explain the classification of Data Mining Systems based on applications.

Data Mining Systems are classified based on their application areas:

1. Healthcare: Predicts diseases and analyzes patient records.
2. Banking and finance: Detects fraud and analyzes financial risks.
3. Retail: Studies customer buying patterns and improves sales.
4. Education: Analyzes student performance and predicts academic results.
5. Telecommunications: Identifies customer behavior and predicts customer churn.
6. E-commerce: Provides personalized product recommendations.

7. Describe the classification of Data Mining Systems based on user interaction.

Based on user interaction, Data Mining Systems can be classified as follows:

1. Query-driven systems: Allow users to specify queries to retrieve particular patterns.
2. Interactive systems: Allow users to interactively explore data and refine mining results.
3. Autonomous systems: Automatically discover patterns with minimal user intervention.

4. Visualization-based systems: Present mining results using charts, graphs, and visual representations.

8. Illustrate a real-world application of a Data Mining System in healthcare.

A real-world application of data mining in healthcare is disease prediction.

- Hospitals collect patient information such as age, symptoms, medical history, and test results.
- Data mining techniques analyze this information to identify disease patterns.
- Classification algorithms can predict whether a patient belongs to a disease-risk category.
- Doctors can use the results to support early diagnosis and treatment planning.

Example: A hospital uses patient records to predict the risk of diabetes and identify patients who may need further medical testing.

9. Define classification in data mining.

Classification is a data mining technique that assigns data objects to predefined classes or categories based on their features.

Example: An email system classifies messages as spam or not spam.

10. Differentiate between classification and clustering in data mining.

Classification	Clustering
It is a supervised learning technique.	It is an unsupervised learning technique.
Uses predefined classes or labels.	Does not require predefined labels.
Assigns data to known categories.	Groups similar data objects together.
Requires labelled training data.	Works mainly with unlabelled data.
Example: Classifying emails as spam or not spam.	Example: Grouping customers based on purchasing behavior.